COMPLETION OF PATTERN

Symmetry, Rotation and Continuity: Finding the Missing Piece

OAVS TGT | Reasoning (Non-Verbal) | Topic 15: sixth non-verbal topic

Syllabus line (Reasoning, 30 Marks): “The Non-Verbal Reasoning topics like Figure Formation, Dice, Triangle, Rule Detection, Images, **Completion of Pattern**, Mirror Images, Figure Matrix and Paper Folding, etc.”

Built from: OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). **Only one completion-of-pattern question appears** in these papers (English 2019 Q56). It is solved with its original figure. The practice questions were generated by program so that each has exactly one option that fits.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	The one PYQ and what it asks
2	Foundation	Types of completion and the checklist method
3	Solved PYQ	English 2019 Q56 with its original figure
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	6 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

One question, in English 2019

Paper	Q. numbers	Count	What was asked
English 2019	56	1	Select the figure which, placed in the blank space of figure X, would complete the pattern
English 2022	none	0	—
Mathematics 2024	none	0	—
Science 2024	none	0	—
Social Studies 2023	none	0	—
Total		**1**	**Answer: Option 2**

PYQ INSIGHT: **Only one question in five papers.** It is a 2 × 2 pattern in which each quadrant has the same ingredients (both diagonals, one arc, a few straight lines) and one quadrant is missing. The answer must supply the arc that finishes the circle and nothing extra. The Figure Matrix questions (Eng 2019 Q53, Eng 2022 Q47) are a different type and have their own PDF.

Pattern	PYQs	Priority
Missing quadrant in a 2 × 2 pattern (symmetry and continuity)	1	★★★ Must
Expected: missing piece by rotation or reflection of the other pieces	0	★★ Expected
Expected: lines that run across the missing region	0	★★ Expected

CHAPTER 2

Foundation

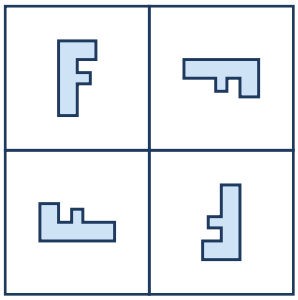
Types and the checklist method

Type	What the pattern does	What the missing piece must do
Rotational pattern	Each quadrant is the previous one turned by 90° (about the centre of the whole figure)	Be the fourth turn of the same motif
Mirror pattern	Quadrants are reflections about the vertical and horizontal middle lines	Be the reflection of its neighbours (horizontal and vertical)
Continuity pattern	Lines and curves run from one quadrant into the next	Join every line that arrives at its edges, and add the closing part of any curve
Repeating motif	The same small pattern appears again and again	Repeat the motif in the right orientation

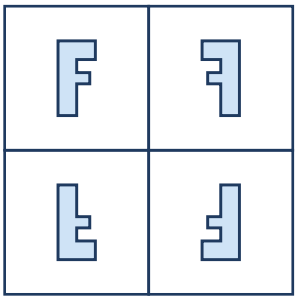
2.1 Checklist

Step	Check
1	**List the ingredients** of the given quadrants (diagonals, arcs, straight lines, small shapes) and count them.
2	**Decide the relation** between quadrants: rotation, reflection or both; test it on two given quadrants.
3	**Apply the relation** to find the missing quadrant.
4	**Check the edges**: every line that ends on an edge of the missing quadrant must meet a line of the neighbour; arcs must join into a smooth curve.
5	**Eliminate** options that have extra pieces (an inner box, an extra arc) or missing pieces.

2.2 The two symmetry patterns (drawn with an F, which has no symmetry of its own)



Rotational: TL, then each quadrant turned a further 90° clockwise (TR, BR, BL).



Mirror: TR mirrors TL left-right; BL mirrors TL up-down; BR is both (a turn of 180°).

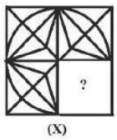
SHORTCUT: Lines at the centre: in a pattern with symmetry, every arc or line that reaches the centre of the whole figure from one quadrant has a partner in each neighbour. Count the lines meeting at the centre in each quadrant; the missing quadrant must give the same count.

CHAPTER 3

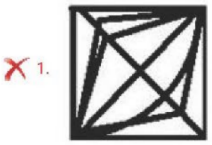
Solved PYQ

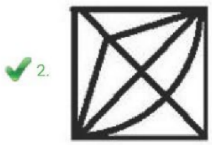
English 2019 Q56

Select a figure from the four alternatives which, placed in the blank space of figure X, would complete the pattern.

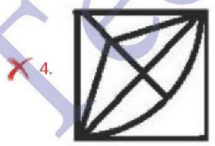


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1. Figure X is a 2 × 2 pattern. Each given quadrant contains **both diagonals** of the quadrant, **one arc**, and a few **straight lines** that fan out from the ends of the arc.
2. The three arcs form **three-quarters of a circle** centred at the middle of the whole figure. The missing quadrant (bottom-right) must supply the fourth quarter: an arc running from the bottom-left corner of that quadrant to its top-right corner, bulging towards the bottom-right.
3. The pattern is symmetric about the middle lines, so the missing quadrant has the same set of ingredients as the others and no more: two diagonals, one arc, the same kind of fan lines.
4. Option 3 has a **small inner square**, which no given quadrant has. Options 1 and 4 have extra arcs or lines on top of the ingredients.
5. Option 2 has the two diagonals, the arc that closes the circle and the fan lines, and nothing extra.

Answer: Option 2

NOTE: The scan is small, so the elimination of options 1 and 4 is read from the figure (extra curves and lines). Option 3 is clear (a small square). The official key is Option 2.

SHORTCUT: **Method:** first find what the given quadrants share (diagonals, one arc, fans). Then find what is **missing from the whole figure** (the last quarter of the circle). The right option has the shared ingredients and the closing arc, with nothing extra.

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Complete the big shape first:** if the pattern shows part of a circle, the missing quadrant supplies the rest.
- **Count ingredients:** if all given quadrants have the same ingredients, the answer has them too and no others.
- **Edges:** every line that ends on an edge must continue in the neighbour.
- **Symmetry:** test rotation and reflection on two given quadrants before applying to the third.
- **Eliminate extras:** an inner box or an extra arc that is not in any quadrant rules the option out.

Trap	How to avoid it
Choosing the busiest option	Extra lines are as wrong as missing lines.
Mixing rotation and reflection	A mirror image is not a rotation for an asymmetric shape.
Ignoring the arc direction	The arc must bulge towards the outer corner so the circle closes.
Forgetting the edges	Lines that run into the missing quadrant must meet lines of the neighbours.

Scope

Related PYQs	Where
Eng 2019 Q53; Eng 2022 Q47 (‘replace the question mark’ in a 3 × 3 matrix)	Figure Matrix (coming)
Mirror images, paper folding	Their own PDFs (coming)

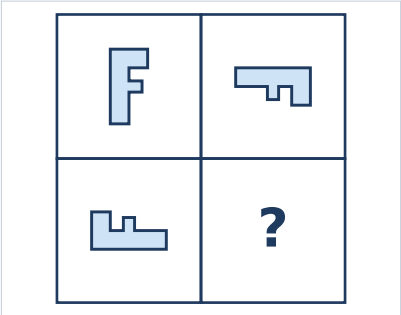
CHAPTER 5

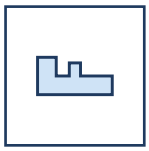
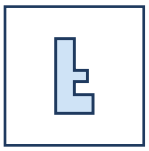
Practice Set

6 original questions; answers on the next page (every option set verified by program)

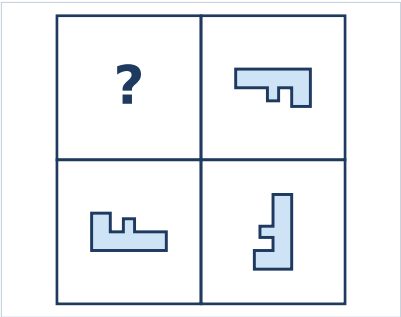
Part A: Rotational pattern

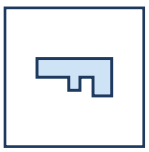
Q1. The pattern is incomplete. Which figure fills the ? so that the pattern is completed?



(A)	(B)	(C)	(D)
			

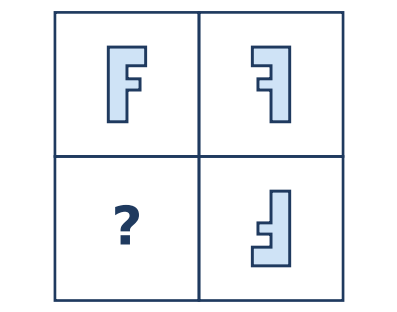
Q2. The pattern is incomplete. Which figure fills the ? so that the pattern is completed?

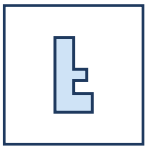


(A)	(B)	(C)	(D)
			

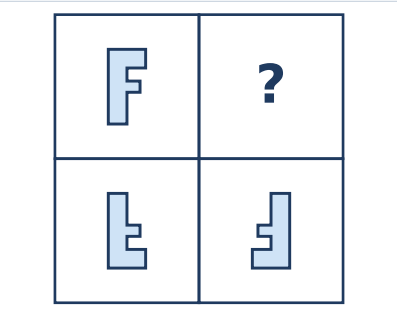
Part B: Mirror pattern

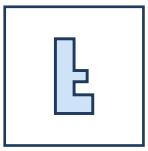
Q3. The pattern is incomplete. Which figure fills the ? so that the pattern is completed?



(A)	(B)	(C)	(D)
			

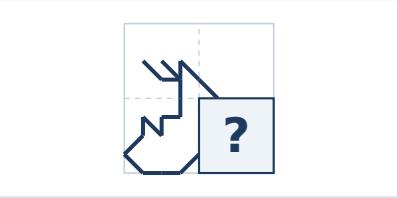
Q4. The pattern is incomplete. Which figure fills the ? so that the pattern is completed?



(A)	(B)	(C)	(D)
			

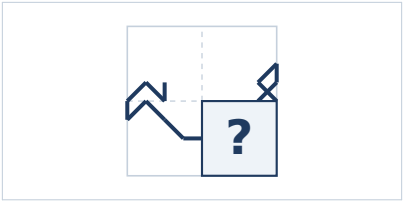
Part C: Lines that must connect (the missing part is the bottom-right quarter)

Q5. Which option, placed in the blank bottom-right quarter, continues the lines of the figure?



(A)	(B)	(C)	(D)
			

Q6. Which option, placed in the blank bottom-right quarter, continues the lines of the figure?



			
(A)	**(B)**	**(C)**	**(D)**

Answer key

Q	Answer	Reason
1	(B)	Each quadrant is the previous one turned 90° clockwise (TL, TR, BR, BL). The missing quadrant (BR) is the motif turned 180°. Wrong options: other turns, or a flipped motif.
2	(B)	Each quadrant is the previous one turned 90° clockwise (TL, TR, BR, BL). The missing quadrant (TL) is the motif turned 0°. Wrong options: other turns, or a flipped motif.
3	(B)	Mirror pattern: TR is TL flipped left-right, BL is TL flipped up-down, BR is TL turned 180°. The missing quadrant is BL.
4	(B)	Mirror pattern: TR is TL flipped left-right, BL is TL flipped up-down, BR is TL turned 180°. The missing quadrant is TR.
5	(C)	Its lines meet the neighbouring lines exactly at the points where they arrive on the top edge and the left edge of the blank quarter; the other options end at different points (checked by program).
6	(C)	Its lines meet the neighbouring lines exactly at the points where they arrive on the top edge and the left edge of the blank quarter; the other options end at different points (checked by program).

CHAPTER 6

Quick Revision

One page for the last day

Type	Method	Check
Rotational pattern	Each quadrant is the previous turned 90°	Missing quadrant = next turn
Mirror pattern	Reflect about the vertical and horizontal middle lines	BR = TL turned 180°
Continuity	Lines meet lines at the edges; arcs join smoothly	Eng 2019 Q56: closing arc, no extras
Elimination	Reject extra inner boxes, extra arcs, missing lines	Option 3 (inner square) is out

ROUTINE: Exam-day routine: list the ingredients; find the missing quarter of the big shape; check edges; eliminate extras. 40 seconds per question. Next in the syllabus order: Mirror Images, Figure Matrix, Paper Folding.